Midterm Project-2025

Deep learning for Image Analysis

Last date of submission:11/11/2025

Dataset: Use the publicly available PCB Defects Dataset**:**
🔗 **https://www.kaggle.com/datasets/akhatova/pcb-defects**

This dataset contains images of printed circuit boards (PCBs) with various types of defects such as ***short circuits***, ***open circuits***, ***missing holes***, and ***mouse bites***.

Tasks

1) Data Preparation & Annotation

- Use **MakeSense.ai** (or *any equivalent tool*) to annotate defect regions either as **bounding boxes** or **masks**.
- Label ***at least four*** categories:
  - ***Short circuit***
  - ***Missing hole***
  - ***Mouse bite***
  - ***Open circuit***
- Export your annotations to **YOLO** or **COCO** format.
- Create a **train/validation/test** split and include screenshots showing:
  - The labeling interface
  - Example labeled images
  - Class counts and distribution summary

2) Model Development

- Train **one** deep learning–based object detection model of your choice:
  - **YOLOv12 / YOLOv8**
  - **Faster R-CNN**
  - **EfficientDet**
- Develop **two pipelines** for comparison:
  1. **Baseline Model**-trained ***without*** preprocessing or augmentation.
  2. **Enhanced Model**-trained ***with*** preprocessing and augmentation techniques.
- Briefly ***explain*** why each chosen method helps improve PCB defect detection.
- Compare the two models and ***discuss*** which performs better and why.

3) Evaluation

- Evaluate both models on the **same test set** using:
  - **Precision**; **Recall**; **F1-score; mAP@0.5**
- Present ***qualitative results*** showing:
  - Predicted bounding boxes with labels and confidence scores
  - Visual comparison between baseline and enhanced models

4) Analysis & Discussion

Discuss key failure cases and possible reasons for them, such as:

- Tiny short circuits near IC pins that are difficult to detect due to low visual contrast or overlapping traces.
- False positives caused by silkscreen lines, solder reflections, or lighting inconsistencies that mimic defect patterns.

Expected Output

- *Annotated dataset screenshots*
- *Model architecture diagram (block schematic)*
- *Training log summary (epochs, loss trends, LR schedule, best epoch)*
- *Inference result images highlighting **short-circuit defects***
- *Performance comparison table (Baseline vs Enhanced)*

Midterm Submission Package Structure, Submission folder/repository should be organized as follows:

```
├── report.pdf                 # Main project report (≤8 pages)
├── code/                      # All scripts for training & inference
│   ├── train.py
│   ├── infer.py
│   ├── eval_metrics.py
│   ├── requirements.txt
│   └── README.md              # Reproduction instructions
│
├── outputs/                   # Visualization and metrics
│   ├── training_curves.png
│   ├── confusion_matrix.png
│   └── sample_predictions/
│       ├── pred_1.jpg
│       ├── pred_2.jpg
│
└── metrics/
    ├── results_baseline.json
    └── results_enhanced.json
```